

1911-1988

MILESTONES

EARLY CAREER

Joins the University of California in 1936; makes key contributions to work with subatomic particles.

INTERNATIONAL ACCLAIM

Wins the Nobel Prize in Physics in 1968 for his discoveries in elementary particle physics.

EXTINCTION THEORY

Works with his son, Walter, in 1980 to propose a theory explaining the extinction of dinosaurs.

Luis Alvarez was a US experimental physicist who received a Nobel Prize for his work with subatomic particles. He was also the first to hypothesize that a meteorite caused the extinction of the dinosaurs.

After obtaining a PhD in physics from Chicago University, Luis Alvarez joined the Radiation Laboratory of the University of California, Berkeley, as an experimental physicist. He specialized in subatomic particles and, in 1937, revealed the process by which a radioactive atom transforms into a new element (K-electron capture). This occurs when the atom's nucleus captures an orbiting electron, which changes the atomic structure and creates a new element. Using a particle accelerator, he also proved that the helium-3 isotope is stable (not radioactive). During World War II, Alvarez developed radar systems and a detonation device for plutonium bombs.

In the late 1950s, he designed a hydrogen bubble chamber, which made it possible to observe and analyze subatomic particles outside the accelerator. This allowed Alvarez and others to discover many new particles and earned him the Nobel Prize in Physics. In 1980, Alvarez theorized that a meteorite led to the extinction of the dinosaurs, after finding iridium (a metal found in meteorites) in rock strata 65 million years old. Support for his theory came a decade later.

Alvarez's dinosaur extinction theory was based on analysis of rock strata from Italy. With the discovery of a vast crater in Mexico in 1990, his theory was confirmed.



LUIS
ALVAREZ